FINAL YEAR PROJECT PROPOSAL

AI-Powered Multi-Modal Presentation Feedback System

Asynchronous Performance Diagnostics Platform

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Executive Summary & Abstract

Core Concept

An asynchronous web platform designed to evaluate and diagnose public speaking and presentation delivery skills automatically.

Multi-Modal Pipeline

Combines Computer Vision (MediaPipe), Acoustic Signal Processing (faster-whisper, Librosa), and Semantic RAG (Gemini API).

Key Deliverable

Generates timestamped performance analytics, objective scoring radar charts, and automated viva defense questions.

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Problem Statement & Motivation

Subjectivity & Bias

Human evaluation varies significantly based on personal mood, attention span, and individual grading preferences.

Lack of Precision

Evaluators cannot accurately measure WPM, pitch stability, eye contact percentage, or subtle hand movement metrics.

Faculty Time Constraints

Mentors and professors lack bandwidth to conduct multiple iterative practice sessions for every student.

Market Fragmentation

Existing platforms cover isolated features (e.g., basic grammar or audio transcript) rather than holistic multi-modal analysis.

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Project Scope & Technical Boundaries

In-Scope Capabilities

- Pre-recorded video (.mp4) and slide deck (.pdf) ingestion
- Eye-contact, posture inclination, and gesture tracking
- Acoustic pacing (WPM), pitch stability, filler words
- Slide-to-speech vector alignment & Viva Q&A generation

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System Architecture Overview

Frontend UI Shell

Built with Next.js & Recharts to deliver an interactive dashboard featuring radar charts, score metrics, and video seeking.

Backend API Layer

Powered by Python FastAPI to handle asynchronous ML processing tasks, payload parsing, and analytical scoring.

Database & Storage

Supabase (PostgreSQL) stores structured JSON metadata while Supabase Blob Storage hosts uploaded media files.

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Computer Vision Engine (MediaPipe)

Gaze & Eye Contact

Leverages 468 Face Mesh landmarks to calculate pupil-to-eye corner vectors for accurate eye contact percentage.

Posture Analysis

Tracks 3D body pose landmarks to calculate shoulder inclination angles and detect slouching during delivery.

Gesture Dynamics

Monitors hand keypoint motion vectors to differentiate between natural gestures, physical rigidity, and fidgeting.

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Acoustic & Speech Processing Engine

Speech Transcription (faster-whisper)

- Word-level timestamp generation
- Precise WPM pacing calculations
- Silent pause tracking (>2.0 seconds)
- Automated filler word detection ('um', 'uh', 'like')

Acoustic Diagnostics (Librosa)

- Short-Time Fourier Transform (STFT) for fundamental pitch (F0) tracking
- Voice monotony identification
- Audio energy thresholding for pauses

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Semantic Analysis & RAG Pipeline

Slide-to-Speech Alignment

- PyMuPDF extracts slide text per page
- Google Gemini Embeddings convert slides and transcript chunks into dense vector representations
- Cosine similarity identifies skipped slides or off-topic speech

Automated Viva Defense Generation

- LangChain orchestrates contextual prompts
- Analyzes speech structural flow
- Google Gemini 1.5 Flash synthesizes 3–5 tailored viva defense questions

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Technology Stack & $0 Deployment Architecture

Layer	Technology Choice	Hosting Platform	Cost
Frontend UI	Next.js + Tailwind CSS	Vercel (Hobby)	$0.00
Backend API	Python FastAPI	Render / Hugging Face	$0.00
Database/Storage	PostgreSQL + Object Storage	Supabase (Free Tier)	$0.00
ML Processing	MediaPipe, faster-whisper, Librosa	Backend Worker	$0.00
LLM & Embeddings	Gemini 1.5 Flash + Embeddings	Google AI Studio	$0.00

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Implementation Roadmap

Phase 1: Setup & Ingestion

Weeks 1–3:
Next.js UI, Supabase schema setup, .mp4 and .pdf upload pipelines.

Phase 2: Core ML Engines

Weeks 4–7:
MediaPipe CV pipeline, faster-whisper transcription, Librosa acoustic scripts.

Phase 3: RAG & Integration

Weeks 8–10:
Slide alignment matching, LangChain viva generation, Next.js dashboard integration.

Phase 4: Optimization & Defense

Weeks 11–12:
End-to-end integration testing, speed optimization, report generation, defense prep.

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Expected Project Outcomes

Unified Diagnostic Dashboard

Displays holistic performance scores broken down across visual, acoustic, and semantic radar charts.

Timestamped Feedback

Clickable error markers on the video timeline allowing presenters to review exact moments of hesitation or slouching.

Zero-Cost Scalability

Completely functional architecture deployed on zero-cost cloud tiers suitable for academic and institutional scaling.

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Questions & Discussion

Thank You

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